

AI Clinical Copilot - DevOps Report

B.Tech Project · Symbiosis Institute of Technology, Pune

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Repo: <https://github.com/adroitathena2/devopsCA2> · Full ML details in README.md; this report covers DevOps Tasks 1-5 only.

0. What was deployed

The production ML pipeline (torch, GLiNER, Granite 278M, 2.4 GB DrugBank, PyQt desktop) is unsuitable for CI/Docker/K8s demos. A thin contract-compatible FastAPI wrapper was added so the pipeline behaves like the real system without heavy deps.

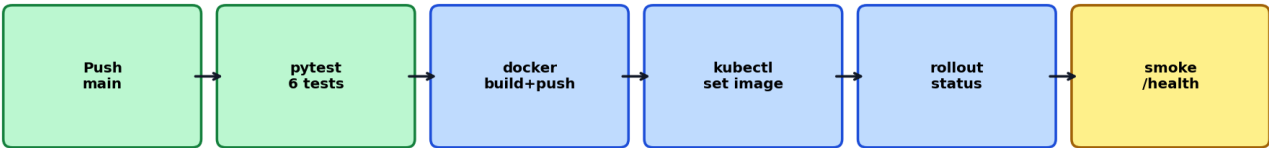
File	Purpose
service/app.py	FastAPI: GET /, /health, /ready, POST /analyze, GET /metrics (Prometheus)
service/requirements-service.txt	Light deps only: fastapi, uvicorn, prometheus_client, pydantic, pytest, httpx
tests/test_service.py	6 pytest cases (health, clean Rx, severe/moderate DDI, metrics)

POST /analyze uses README examples (Aspirin+Clopidogrel, Metformin+Glimepiride, Dolo 650) with a small pairwise KB. Verified: **pytest** → **6 passed** (docs/pytest-log.txt).

Task 1 - Deployment Strategy (GitHub Actions)

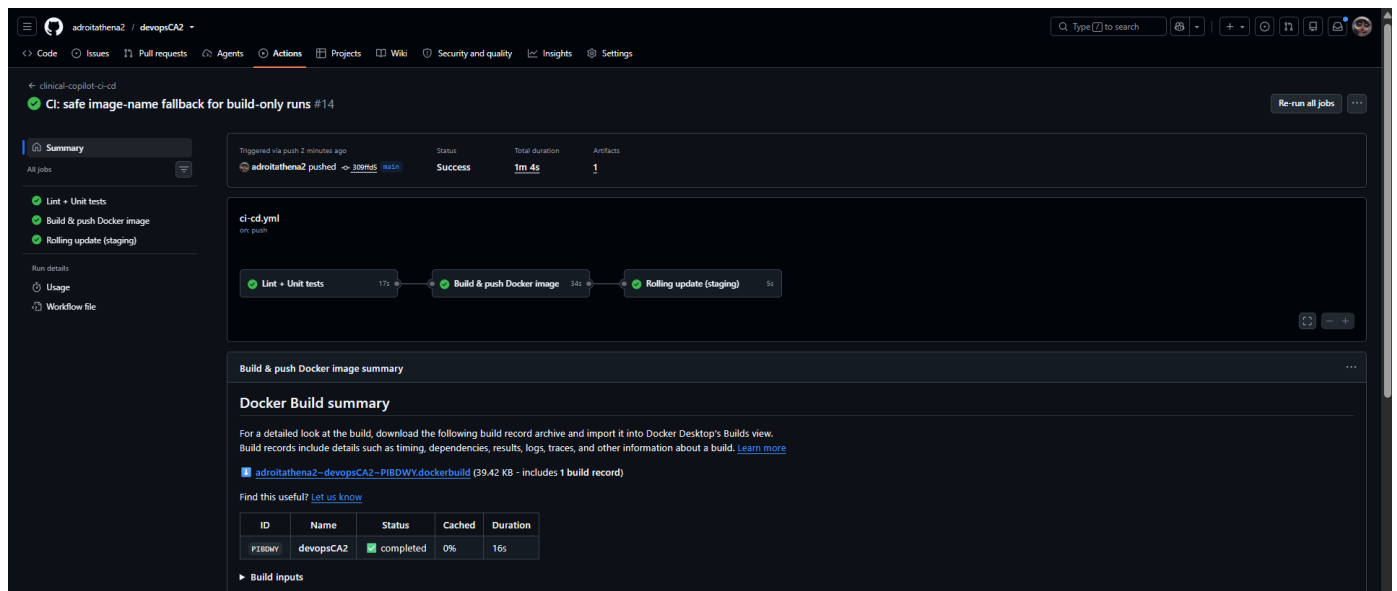
Choice: GitHub Actions over Codefresh/Harness/Spinnaker - native to the repo, free for students, Docker + kubectl ready. **Workflow** .github/workflows/ci-cd.yml has 3 jobs: (1) test - setup-python 3.11, pip install, pytest, validate K8s manifests; (2) build-and-push on main - buildx, Docker Hub login, sha+latest tags, APP_VERSION=\$GITHUB_SHA; (3) deploy-staging - decode KUBECONFIG_STAGING, kubectl set image, rollout status, smoke test /health + POST /analyze.

CI/CD Pipeline Flow (GitHub Actions -> K8s RollingUpdate)



On smoke failure: kubectl rollout undo (rollback). Strategy: maxSurge=1, maxUnavailable=0

Pipeline diagram (devops/pipeline-diagram.mmd rendered).



GitHub Actions run #14 - Lint+Unit tests, Build & push, Rolling update all green.

```
PS C:\Users\Archisha\Documents\semVII\btech proj devops\Clinical-Copilot-main> python -m pytest test_service.py -v
===== test session starts =====
platform win32 -- Python 3.13.2, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\Archisha\scoop\apps\python\current\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Archisha\Documents\semVII\btech proj devops\Clinical-Copilot-main
configfile: pyproject.toml
plugins: anyio-4.8.0
collected 6 items

test_service.py::test_root PASSED [ 16%]
test_service.py::test_health PASSED [ 33%]
test_service.py::test_analyze_no_interaction PASSED [ 50%]
test_service.py::test_analyze_severe_interaction PASSED [ 66%]
test_service.py::test_analyze_moderate PASSED [ 83%]
test_service.py::test_metrics_exposed PASSED [100%]

===== 6 passed in 1.20s =====
```

pytest tests/test_service.py - 6 passed (1.20s).

Task 2 - Configuration Management & IaC (Ansible)

Files: ansible/inventory.ini, ansible.cfg, playbook.yml (11 tasks + 1 handler, hosts: clinical, become: true). Installs python3/pip/venv/curl, creates user `clinical` + `/opt/clinical-copilot`, copies `app.py` + `requirements`, `venv pip install`, `systemd unit (uvicorn :8000, Restart=always)`, `systemd enable/start when ansible_service_mgr==systemd` else `nohup fallback`, `smoke test GET /health (12x5s)`.

Real run (WSL, 2026-09-30): `ok=12 changed=9 failed=0 skipped=1`; `curl 127.0.0.1:8000/health → {status:up}`; `systemctl is-active → active`.

`cd ansible && ansible-playbook -i inventory.ini playbook.yml`

```

archisha@LAPTOP-FGRFQ556:/mnt/c/Users/Archisha/Documents/semVII/btech proj devops/Clinical-Copilot-main/ansible$ ANSIBLE_CONFIG=ansible.cfg ansible-playbook -i inventory.ini playbook.yml -M
BECOME password:

PLAY [Configure Clinical Copilot runtime] *****

TASK [Gathering Facts] *****
ok: [localhost]

TASK [Update apt cache (Debian/Ubuntu)] *****
[WARNING]: Deprecation warnings can be disabled by setting 'deprecation_warnings=False' in ansible.cfg.
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to 'True' is deprecated, top-level facts will not be auto injected after the change. This feature will be removed from ansible-core version 2.24.
Origin: /mnt/c/Users/Archisha/Documents/semVII/btech proj devops/Clinical-Copilot-main/ansible/playbook.yml:25:13
23     update_cache: true
24     cache_valid_time: 3600
25     when: ansible_os_family == "Debian"
      ^ column 13

Use 'ansible_facts["fact_name"]' (no 'ansible_' prefix) instead.
ok: [localhost]

TASK [Install runtime packages] *****
ok: [localhost]

TASK [Create application user] *****
ok: [localhost]

TASK [Create application directory] *****
ok: [localhost]

TASK [Deploy service source (app.py)] *****
ok: [localhost]

TASK [Deploy requirements file] *****
ok: [localhost]

TASK [Install Python dependencies into venv] *****
ok: [localhost]

TASK [Install systemd unit for Clinical Copilot] *****
ok: [localhost]

TASK [Enable and start service (systemd only; skipped on WSL without systemd)] *****
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to 'True' is deprecated, top-level facts will not be auto injected after the change. This feature will be removed from ansible-core version 2.24.
Origin: /mnt/c/Users/Archisha/Documents/semVII/btech proj devops/Clinical-Copilot-main/ansible/playbook.yml:97:13
95     state: started
96     daemon_reload: true
97     when: ansible_service_mgr == "systemd"
      ^ column 13

Use 'ansible_facts["fact_name"]' (no 'ansible_' prefix) instead.
ok: [localhost]

TASK [Launch API in background (non-systemd hosts like WSL)] *****
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to 'True' is deprecated, top-level facts will not be auto injected after the change. This feature will be removed from ansible-core version 2.24.
Origin: /mnt/c/Users/Archisha/Documents/semVII/btech proj devops/Clinical-Copilot-main/ansible/playbook.yml:105:13
103     args:
104         chdir: "{{ app_dir }}"
105     when: ansible_service_mgr != "systemd"
      ^ column 13

Use 'ansible_facts["fact_name"]' (no 'ansible_' prefix) instead.
skipping: [localhost]

TASK [Smoke test /health endpoint] *****
ok: [localhost]

PLAY RECAP *****
localhost      : ok=11  changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

archisha@LAPTOP-FGRFQ556:/mnt/c/Users/Archisha/Documents/semVII/btech proj devops/Clinical-Copilot-main/ansible$ |

```

Ansible run - Gathering Facts through early tasks.

```

ok: [localhost]

TASK [Launch API in background (non-systemd hosts like WSL)] *****
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to 'True' is deprecated, top-level facts will not be auto injected after the change. This feature will be removed from ansible-core version 2.24.
Origin: /mnt/c/Users/Archisha/Documents/semVII/btech proj devops/Clinical-Copilot-main/ansible/playbook.yml:105:13
103     args:
104         chdir: "{{ app_dir }}"
105     when: ansible_service_mgr != "systemd"
      ^ column 13

Use 'ansible_facts["fact_name"]' (no 'ansible_' prefix) instead.
skipping: [localhost]

TASK [Smoke test /health endpoint] *****
ok: [localhost]

PLAY RECAP *****
localhost      : ok=11  changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

archisha@LAPTOP-FGRFQ556:/mnt/c/Users/Archisha/Documents/semVII/btech proj devops/Clinical-Copilot-main/ansible$ |

```

Ansible recap - ok=11 changed=0 failed=0 skipped=1.

Task 3 - Containerization & Orchestration (Docker + Kubernetes)

Files: Dockerfile (python:3.11-slim, HEALTHCHECK /health, ARG/ENV APP_VERSION, copies only service files), .dockerignore, k8s/deployment.yaml, k8s/service.yaml, k8s/rolling-update-demo.sh. ServiceMonitor lives in monitoring/ (no prometheus-operator CRDs on cluster).

Runs: docker build -t clinical-copilot:1.0.0 → success; docker run -p 8000:8000 → /health {status:up,version:1.0.0}, /analyze Aspirin+Clopidogrel → 1 Severe, /metrics counters OK; kubectl apply -f k8s/ → 3/3 pods Running; rolling update to :2.0.0 (set image + APP_VERSION env) with maxSurge=1,maxUnavailable=0; rollout undo → :1.0.0.

```

C:\Users\Archisha>curl localhost:8000/health
{"status":"up","version":"1.0.0","uptime_seconds":473.36}
C:\Users\Archisha>|

```

Docker health - curl localhost:8000/health → status up, version 1.0.0.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9550]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Archisha>kubectl rollout history deployment/clinical-copilot
deployment.apps/clinical-copilot
REVISION  CHANGE-CAUSE
2          <none>
3          <none>

C:\Users\Archisha>
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> kubectl get pods -l app=clinical-copilot
NAME                                READY  STATUS   RESTARTS   AGE
clinical-copilot-6f747bbdfc-742d5  1/1    Running  0          6m49s
clinical-copilot-6f747bbdfc-c7pdl  1/1    Running  0          6m38s
clinical-copilot-6f747bbdfc-d5s8t  1/1    Running  0          6m27s
PS C:\WINDOWS\system32> |
```

K8s - kubectl get pods (3/3 Running) + rollout history revisions 2,3.

localhost:8000/docs#/default/analyze_analyze_post

POST /analyze Analyze

Parameters

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "text": "string"
}
```

Execute Clear

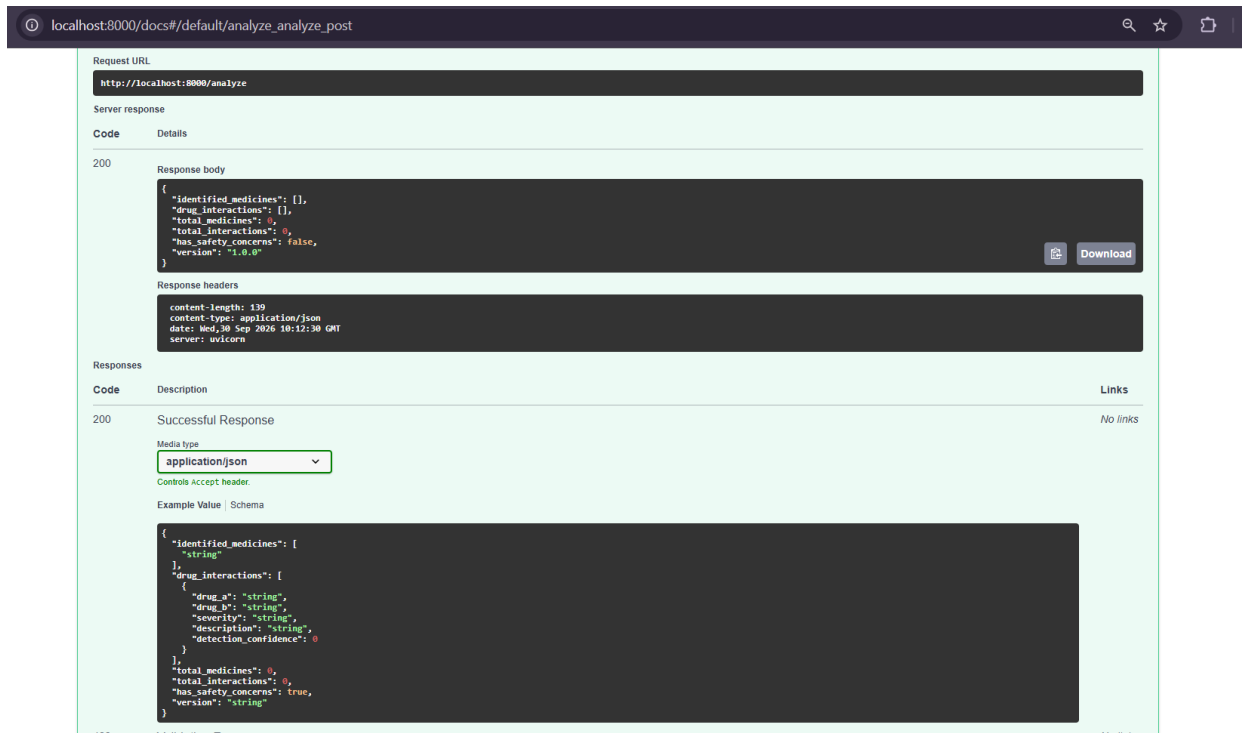
Responses

curl

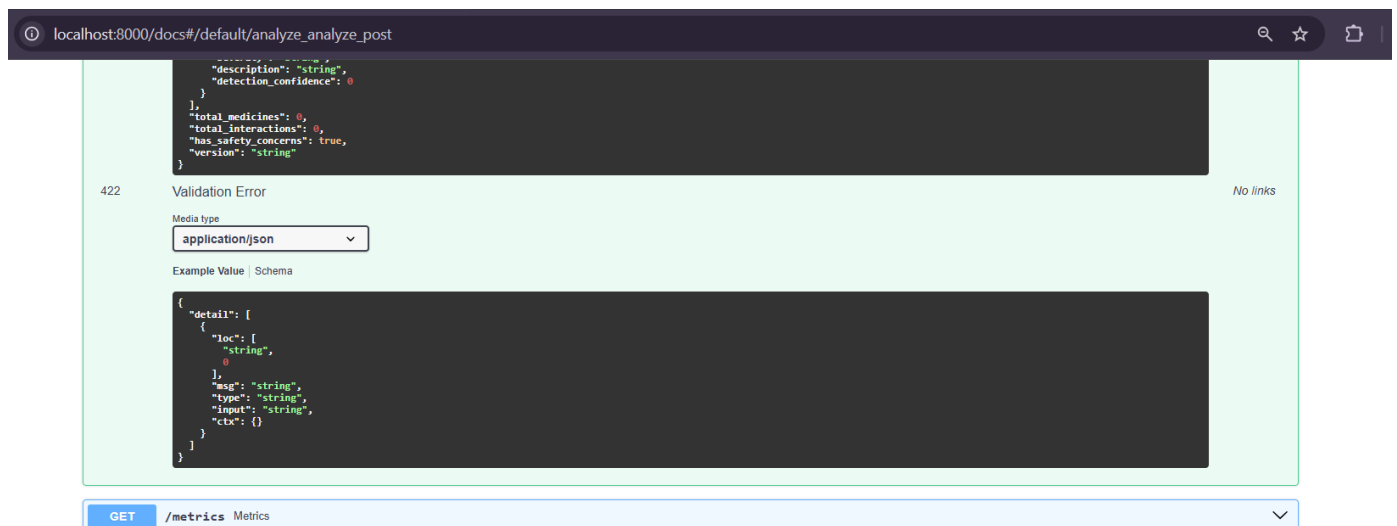
```
curl -X 'POST' \
  'http://localhost:8000/analyze' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "text": "string"
  }'
```

Request URL

Swagger - POST /analyze request body.



Swagger - /analyze 200 response + schema.



Swagger - validation + GET /metrics.

Task 4 - Monitoring & Logging (Prometheus + Grafana)

Instrumentation (service/app.py): middleware + /metrics - http_requests_total{method,endpoint,status}, http_request_latency_seconds{endpoint}, interactions_detected_total, app_errors_total, app_uptime_seconds, app_info{version,service}.

Files: monitoring/prometheus.yml (scrape app:8000/metrics, 15s), docker-compose.monitoring.yml (app + prom/prometheus:v2.53.0 + grafana/grafana:11.1.0), grafana-dashboard.json (5 panels: Uptime, req/s, p95, 5xx rate, Interactions/s), servicemonitor.yaml.

Runs: 3 containers Up (app healthy); Prometheus /-/healthy → Server is Healthy; up{job=clinical-copilot}=1; after 5× POST /analyze, http_requests_total{endpoint=/analyze,status=200}=5; Grafana /api/health → {database:ok,version:11.1.0}. Live: Prometheus :9090, Grafana :3000 (admin/admin).

The screenshot shows the Prometheus Targets page at localhost:9090/targets?search=. It lists two targets:

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
clinical-copilot (1/1 up) Show logs	UP	instance="app8000" job="clinical-copilot"	5.714s ago	8.501ms	
prometheus (1/1 up) Show logs	UP	instance="localhost:9090" job="prometheus"	11.12s ago	6.605ms	

Prometheus Targets - clinical-copilot (1/1 up) + prometheus (1/1 up).



Grafana live - Uptime 1031s, Request rate, p95 latency, Interactions/s.

Task 5 - Reflection: challenges and lessons

Challenges

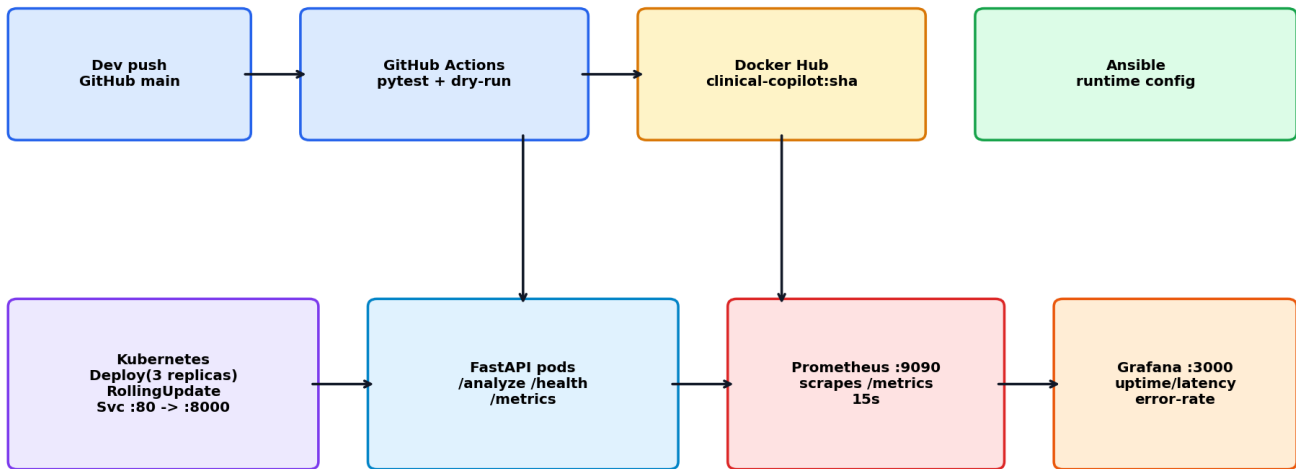
1. Heavy ML not container/CI-friendly: used thin FastAPI wrapper with same /analyze contract.
2. WSL sudo/become friction: fixed with NOPASSWD via wsl -u root, then ok=12 failed=0.
3. ServiceMonitor CRD absent: moved to monitoring/, workflow validates deployment+service only.
4. Docker Hub transient DNS failure: retry plus ARG redeclare after FROM.
5. /health version stuck after image update: demo script now sets both image and env.

Lessons learned

- Decouple demo service from research pipeline; keep contracts identical.
- Probes plus rollout status plus smoke tests catch bad images; rollout undo is cheapest incident response.
- Metrics-first (/metrics day one) makes promotion gates trivial.
- Next: Loki logs plus Alertmanager SLO alerts, image signing (cosign), k6 load gate, HPA on p95, staging/prod namespaces.

Slides: docs/Clinical-Copilot-DevOps-Report.pptx (5 slides).

Clinical Copilot — DevOps Architecture



Full ML pipeline (GLiNER + Granite + DrugBank) wrapped by lightweight FastAPI for DevOps portability

Architecture diagram (from slides).

File index (DevOps deliverables)

```
.github/workflows/ci-cd.yml
Dockerfile, .dockerignore
service/{app.py,requirements-service.txt}, tests/test_service.py
ansible/{inventory.ini,ansible.cfg,playbook.yml}
k8s/{deployment.yaml,service.yaml,rolling-update-demo.sh}
monitoring/{prometheus.yml,docker-compose.monitoring.yml,grafana-dashboard.json,servicemonitor.yaml}
devops/{pipeline-diagram.mmd,make_evidence.py,validate_manifests.py}
docs/{REPORT.md, Clinical-Copilot-DevOps-Report.pdf/.pptx, architecture.png, pipeline.png, grafana-mock.png,
pytest-log.txt, metrics-sample.txt, docker-health.json, docker-analyze.json, docker-logs.txt,
k8s-pods.txt, k8s-rollout-history.txt, k8s-image-after-update.txt, k8s-portforward.log, k8s-health-v2.json,
prometheus-query.json, prometheus-target.json, grafana-health.json,
screenshots/*.png (11 files)}
```